Inverse Hyperbolic Functions.

If $x = \sinh y$ and $y = \sinh^{-1} x$, then we say that y is the inverse hyperbolic sine of x. Illy we can define $\cosh^{-1} x$, $\tanh^{-1} x$ etc.

Q. If x is real, S.T $\sinh^{-1} x = \log x + \sqrt{x^2+1}$?

Let $\sinh^{-1} x = y$

$$x = \sinh y = \frac{e^{y} - e^{-y}}{2}$$

$$2x = e^{y} - e^{-y} = e^{y} - 1/e^{y}$$

$$2xe^{y} = e^{2y} - 1$$

$$(e^{y})^2 - 2xe^{y} - 1 = 0.$$

$$e^{y} = \frac{2x \pm \sqrt{4x^2+4}}{2} = x \pm \sqrt{x^2+1}$$

Since e^y is a +'ve quantity, taking log

$$y = \log\left(x+\sqrt{x^2+1}\right)$$

$$\sinh^{-1}x = \log\left(x+\sqrt{x^2+1}\right)$$

Q. S.T $\cosh^{-1}x = \log\left(x+\sqrt{x^2-1}\right)$?

Let $\cosh^{-1}x = y$

$$x = \cosh y$$

$$= \frac{e^y + e^{-y}}{2}$$

$$2x = e^y + 1/e^y$$

$$2xe^y = (e^y)^2+1$$

$$(e^y)^2 - 2xe^y + 1 = 0.$$

$$e^y = \frac{2x \pm \sqrt{4x^2-4}}{2}$$

$$= x \pm \sqrt{x^2-1}$$

Since e^y is +'ve, $e^y = x+\sqrt{x^2-1}$

$$y = \log\left(x+\sqrt{x^2-1}\right)$$

$$\cosh^{-1}x = \log\left(x+\sqrt{x^2-1}\right)$$

Q S.T $\tanh^{-1} x = \frac{1}{2} \log\left(\frac{1+x}{1-x}\right)$?

Let $\tanh^{-1} x = y$

$x = \tanh y$

$= \frac{e^{y} - e^{-y}}{e^{y} + e^{-y}}$

$$\frac{1+x}{1-x} = \frac{1 + e^{y} - e^{-y} / e^{y} + e^{-y}}{1 - e^{y} - e^{-y} / e^{y} + e^{-y}}$$

$$= \frac{e^{y} + \cancel{e^{-y}} + e^{y} - \cancel{e^{-y}}}{\cancel{e^{y}} + e^{-y} - \cancel{e^{y}} + e^{-y}} = \frac{2e^{y}}{2e^{-y}} = e^{2y}$$

$$e^{2y} = \frac{1+x}{1-x}$$

$$2y = \log\left(\frac{1+x}{1-x}\right)$$

$$y = \frac{1}{2} \log\left(\frac{1+x}{1-x}\right)$$

$$\tanh^{-1} x = \frac{1}{2} \log\left(\frac{1+x}{1-x}\right)$$

Q. Separate into real and imaginary parts $\sin^{-1}(\cos\theta + i\sin\theta)$ where θ is real?

Let $\sin^{-1}(\cos\theta + i\sin\theta) = x+iy$

$$\cos\theta + i\sin\theta = \sin(x+iy)$$
$$= \sin x \cos(iy) + \cos x \sin(iy)$$
$$= \sin x \cosh y + i\cos x \sinh y$$

Equating real and imaginary,

$$\cos\theta = \sin x \cosh y \quad \text{——— (1)}$$
$$\sin\theta = \cos x \sinh y \quad \text{——— (2)}$$

squaring and adding (1) and (2)

$$\cos^2\theta + \sin^2\theta = \sin^2 x \cosh^2 y + \cos^2 x \sinh^2 y$$
$$1 = \sin^2 x(1+\sinh^2 y) + \cos^2 x \sinh^2 y$$
$$= \sin^2 x + \sin^2 x \sinh^2 y + \cos^2 x \sinh^2 y$$
$$= \sin^2 x + \sinh^2 y(\sin^2 x + \cos^2 x)$$
$$= \sin^2 x + \sinh^2 y$$

$$\sinh^2 y = 1 - \sin^2 x = \cos^2 x$$

$$\cos^2 x = \sinh^2 y$$
$$= \frac{\sin^2\theta}{\cos^2 x} \quad (\text{from (2)}) \quad \sinh y = \frac{\sin\theta}{\cos x}$$

$$\cos^4 x = \sin^2\theta$$
$$\cos^2 x = \sin\theta$$
$$\cos x = \pm\sqrt{\sin\theta}$$
$$x = \cos^{-1}\sqrt{\sin\theta} \quad \text{——— (3)}$$

From ②, $\sinh y = \dfrac{\sin\theta}{\cos x}$

$$= \frac{\sin\theta}{\sqrt{\sin\theta}} \quad \text{from ③.}$$

$$\sinh y = \sqrt{\sin\theta}.$$

$$\frac{e^{y} - e^{-y}}{2} = \sqrt{\sin\theta}$$

$$e^{y} - 1/e^{y} = 2\sqrt{\sin\theta}$$

$$(e^{y})^2 - 1 = 2e^{y}\sqrt{\sin\theta}$$

$$(e^{y})^2 - 2e^{y}\sqrt{\sin\theta} - 1 = 0$$

$$e^{y} = \frac{2\sqrt{\sin\theta} \pm \sqrt{4\sin\theta + 4}}{2}$$

$$e^{y} = \sqrt{\sin\theta} \pm \sqrt{\sin\theta + 1}$$

$$\Rightarrow y = \log\left(\sqrt{\sin\theta} \pm \sqrt{\sin\theta + 1}\right)$$